INCH-POUND

MIL-C-70590B (AR)
3 April 1991
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

CONTROL ASSEMBLY TEST PANEL TS-4215/TSM-100B: 9361261

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Control Assembly Test Panel, hereinafter referred to as the "test set," which forms a part of the Electronics System Test Set AN/TSM-100B (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, U.S Army ARDEC, ATTN SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-2000	-	Standard Requirements for Soldered Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9361260	Schematic Diagram, Electrical, Control Panel, Test Unit
9361261	Panel Assembly, Control Test Unit

SPECIAL PACKAGING INSTRUCTION

PS9361261	- Packaging of Test Set, Control Panel

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9361261, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the test set shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The test set shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 10-55-10 hertz sweep time is 3 minutes 45 seconds ± 10%.

3.3.2 Shock. The test set shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's + 10%, 11 milliseconds + 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The test set shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The test set shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The test set shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 deg F and +155 deg F.

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3.3.4.2 Non-operating (storage). The test set shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg F for a period of 24 hours and +160 deg F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The test set shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the test set.

3.5 Performance.

3.5.1 Switch functions. With the SWITCH FUNCTION CHECK switch and the CONTINUITY CHECK switch in the positions specified in Table I (Switch functions) and with a +24.0 ± 0.1 Vdc power source limited to 1 ampere applied between TP5 (+) and TP4 (ground), there shall be +24.0 ± 0.1 Vdc between the pins specified in Table I and GROUND. The +24 Vdc POWER indicator lamp shall LIGHT when the SWITCH FUNCTION CHECK switch is in position B and the CONTINUITY CHECK switch is in position 7.

TABLE I. Switch functions.

SWITCH FUNCTION CHECK SWITCH POSITION	CONTINUITY CHECK SWITCH POSITION	OUTPUT VOLTAGE AT CONNECTOR AND PIN
A	1	J23-B J24-A
A	2	J24-G J24-K
A	3	J24-C J24-L
A	4	No connection
A	5	J22-BT J22-BM
A	6	J22-BU J22-BP
A	7	J22-BS J22-BV
A	8	J22-BA

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TABLE I. Switch functions (continued).

SWITCH FUNCTION CHECK SWITCH POSITION	CONTINUITY CHECK SWITCH POSITION	OUTPUT VOLTAGE AT CONNECTOR AND PIN
A	9	J22-AZ J22-AP
A	OFF	J22-AR J22-BC
B	1	J22-AV J22-AX
B	2	J23-X J23-W J22-AU J22-AL J22-BB J22-BJ J22-i J22-k
B	3	J22-AY
B	4	J22-AD
B	5	J22-AH
B	6	J23-V J22-BL J22-BN J22-u J22-t
B	7	J22-AW
B	8	J22-j
B	9	J22-w J22-AE J22-y J22-z
B	OFF	J22-x J22-AJ
C	1	J22-m J22-q
C	2	J22-AF

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TABLE I. Switch functions (continued).

SWITCH FUNCTION CHECK SWITCH POSITION	CONTINUITY CHECK SWITCH POSITION	OUTPUT VOLTAGE AT CONNECTOR AND PIN
C	3	J22-AT J22-BD J23-R J23-h J23-j
C	4	J22-v
C	5	J22-BR
C	6	J23-F
C	7	J22-d
C	8	J22-h J22-c
C	9	J23-Z
C	OFF	J23-n
D	1	J23-P
D	2	J23-k J23-N
D	3	J23-c
D	4	J23-s (See Note) J23-A
D	5	J23-b

NOTE: Leakage current through blocking diode to J23-S shall not exceed 200 microampere.

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TABLE I. Switch functions (continued).

SWITCH FUNCTION CHECK SWITCH POSITION	CONTINUITY CHECK SWITCH POSITION	OUTPUT VOLTAGE AT CONNECTOR AND PIN
D	6	J22-AK
D	7	J22-AS J23-m
D	8	J22-p
D	9	J23-e
D	OFF	J23-h J23-j J23-R
E	1	J23-d J23-E
E	2	J23-r
E	3	J23-g J23-C
E	4	J31-A
E	5	J31-L
E	6	J31-N
E	·J	J31-S
E	8	J31-E
E	9	J31-J
E	OFF	J31-R
F	1	J31-U
F	2	J31-C
F'	3	J31-T
F	4	J31-B
F	5	J31-X
F	6	J31-Y

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TABLE I. Switch functions (continued).

SWITCH FUNCTION CHECK SWITCH POSITION	CONTINUITY CHECK SWITCH POSITION	OUTPUT VOLTAGE AT CONNECTOR AND PIN
F	7	J31-Z
F	8	J31-W
F	9	J31-V
F	OFF	J31-c
G	1	J31-G
G	2	J31-b
G	3	J31-M
G	4	J31-K
G	5	J31-P
G	6	J31-F
G	7	J31-H
G	8	J31-D
G	9	J31-a
G	OFF	J22-AC

3.5.1.1 SWITCH FUNCTION GROUND. With the SWITCH FUNCTION CHECK switch in position B and the CONTINUITY CHECK switch in position 6, the resistance between TP5 and TP4 (ground) shall be less than 2.5 ohms. With the SWITCH FUNCTION CHECK switch in position D and the CONTINUITY CHECK switch in POSITION 9, the resistance between TP5 and TP4 (ground) shall be less than 2.5 ohms.

3.5.2 SWITCH FUNCTION CHECK indicator. With +22.0 ± 0.1 Vdc applied between the specified pin in Table II (Switch function check indicator) and GROUND, the SWITCH FUNCTION CHECK indicator lamp shall LIGHT only when the SWITCH FUNCTION CHECK switch is in the position specified in Table II.

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TABLE II. Switch function check indicator.

SWITCH FUNCTION CHECK SWITCH POSITION	VOLTAGE APPLIED TO CONNECTOR AND PIN
A	J23-r
B	J23-p
C	J23-d
D	J23-m
E	J22-AM
F	J23-b
G	J22-j
H	J22-AF
I	J23-c

3.5.3 CONTINUITY CHECK indicator. After +22.0 + 0.1 Vdc is applied between the specified pin in Table III (Continuity check indicator) and GROUND, the CONTINUITY CHECK indicator lamp shall light only when the CONTINUITY CHECK switch is in the position specified in Table III.

TABLE III. Continuity check indicator.

CONTINUITY CHECK SWITCH POSITION	VOLTAGE APPLIED TO CONNECTOR AND PIN
2	J22-BT
3	J22-BU
4	J22-BS
5	J22-AZ
6	J22-AR
7	J22-BD
8	J22-a
9	J22-h

3.5.4 POTENTIOMETER SELECT switch.

3.5.4.1 TEST POINT NO. 1 (TP1). With +24.0 ± 0.1 Vdc applied between TEST POINT NO. 1 (TP1) and GROUND, there shall be +24.0 ± 0.1 Vdc between the pins specified in Table IV (Test point no. 1) and GROUND when the POTENTIOMETER SELECT switch is in the position specified in Table IV.

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TABLE IV. Test point no. 1.

POTENTIOMETER SELECT SWITCH POSITION	OUTPUT VOLTAGE AT CONNECTOR AND PIN
1	J23-f
2	J23-f
3	J24-F
4	J23-M
5	J23-M
6	J22-n

3.5.4.2 TEST POINT NO. 2 (TP2). With +24.0 ± 0.1 Vdc applied between TEST POINT NO. 2 (TP2) and GROUND, there shall be +24.0 ± 0.1 Vdc between the pins specified in Table V (Test point no. 2) and GROUND when the POTENTIOMETER SELECT switch is in the position specified in Table V.

TABLE V. Test point no. 2.

POTENTIOMETER SELECT SWITCH POSITION	OUTPUT VOLTAGE AT CONNECTOR AND PIN
1	J23-U
2	J23-U
3	J24-D
4	J23-f
5	J23-f
6	J22-r

3.5.4.3 TEST POINT WIPER (TP3). With +24.0 ± 0.1 Vdc applied between TEST POINT WIPER (TP3) and GROUND, there shall be +24.0 ± 0.1 Vdc between the pins specified in Table VI (Test point wiper) and GROUND when the POTENTIOMETER SELECT switch is in the position specified in Table VI.

TABLE VI. Test point wiper.

POTENTIOMETER SELECT SWITCH POSITION	OUTPUT VOLTAGE AT CONNECTOR AND PIN
1	J23-S
2	J23-T
3	J23-p
4	J23-P
5	J23-D

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3.5.5 RC network.

3.5.5.1 Conditions of operation. The requirements of 3.5.5 shall be met with the conditions listed below:

a. +24.0 + 0.05 Vdc (600 milliamperes or greater) source applied between J114-C(+) and J114-A(-).

b. A 1 Kohm ± 5 percent, 1 watt, load applied between J22-AY and J22-BL.

c. +24 Vdc POWER switch in the "ON" position. The +24 Vdc POWER indicator lamp shall be "LIT."

3.5.5.2 RC charging. The waveform shown in Figure 1 shall be obtained across terminals E3 and E4 when the POTENTIOMETER SELECT switch is placed in position 7.

3.5.5.3 RC discharging. The waveform shown in Figure 2 shall be obtained across terminals E1(+) and E2 when the POTENTIOMETER SELECT switch is switched from position 7 to position 6. The waveform shown in Figure 2 shall also be obtained across J22-AY(+) and J22-BL when the POTENTIOMETER SELECT switch is switched from position 7 to position 8.

3.5.6 RELAY TEST switch. With the RELAY TEST switch in the "ON" position, the resistance between J23-a and TP4 (ground) shall be less than 2.5 ohms.

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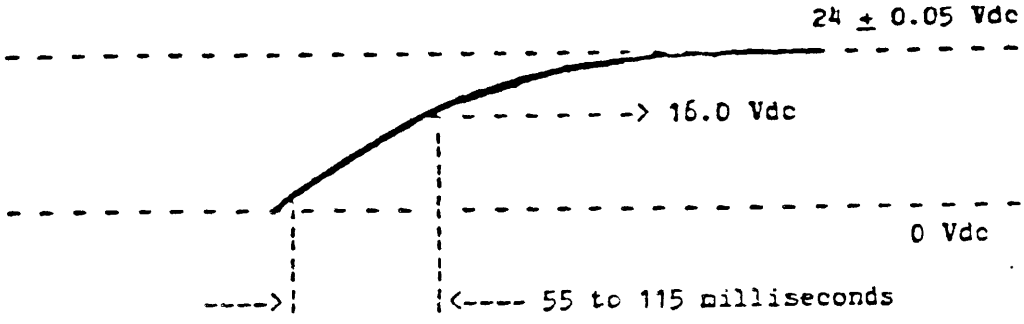


FIGURE 1. RC charging waveform.

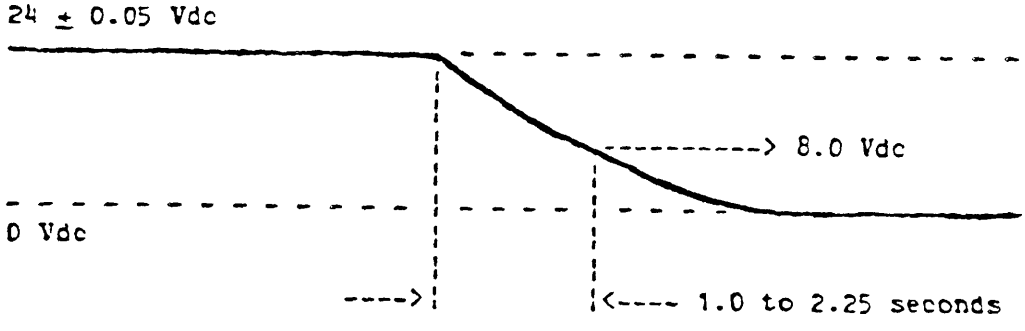


FIGURE 2. RC discharging waveform.

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3.5.7 15 V regulater circuit. The 15 V regulator circuit shall provide an adjustable dc voltage from 4 to 20 Vdc at J23-J and J23-f when compared to TP4 (ground) .

3.5.8 Blocking Diode Continuity. Voltage drop between P3-s (+) and P3-A, at a current flow of 10 milliamperes, shall not exceed 1.0 volt.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5 and all applicable drawings. The inspection set forth in this specification shall become part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified. all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg to +90 deg F
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) test-sets for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table VII (fictional Test) and Table VIII (Special Sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term “inspection lot” is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

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4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table VII shall be performed on a 100% basis.

TABLE VII. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

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4.5 Special sampling. One test set out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table VII and shall then meet the tests in Table VIII. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE VIII. Special sampling.

CHARACTERISTICS	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be-suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the “Methods of Inspection” specified in 4.7 of-this specification and with the component inspection procedures specified in “Examination” and “Test Facilities” requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment

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is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table IX (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE IX. Test equipment list.

ITEM NUMBER		DESCRIPTION
1.	Ohmmeter	o to 1000 ohms. Accuracy and resolution 5 percent or better.
2.	Digital voltmeter (DVM)	A range of at least .0010 to 999.9 Vdc with four significant figures. Accuracy ± 0.2 percent of reading. Input impedance 10 megohms or better.
3.	Power supply	0-30 Vdc and O-1 ampere, current and voltage adjustable, regulated to ± 1 percent or better. Less than 0.1 percent ripple.
4.	Oscilloscope	Dc -15 MHz; input impedance 10 megohms or greater, 5 volt/cm sensitivity; internal sweep; 10 milliseconds/cm and 200 millisecond sweep speeds.
5.	Resistor	1 Kohm ± 5 percent, 1 watt.

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

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4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The test set shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the test set shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the test set shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating) This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating. This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

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4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance.

4.7.3.1 SWITCH FUNCTION CHECK and CONTINUITY CHECK test. Connect the power supply (item 3, Table IX) between TP5(+) and TP4 (ground). Set the power supply current adjust to 800 mA. Using the DVM (item 2, Table IX) to monitor the power supply output voltage, set the voltage to 24.0 ± 0.1 Vdc. Set the SWITCH FUNCTION CHECK and CONTINUITY CHECK rotary switches to the positions of Table I. Using the DVM, with its (-) terminal at GROUND, verify the output of 24.0 ± 0.1 Vdc at the connector and pins specified in Table I for each specified combination of switch settings. With the SWITCH FUNCTION CHECK switch at position B and CONTINUITY CHECK switch at position z, verify that the +24 Vdc POWER indicator lamp is "LIT ."

4.7.3.2 SWITCH FUNCTION ground test. Using the ohmmeter (item 1, Table IX), verify the requirements of 3.5.1.1.

4.7.3.3 SWITCH FUNCTION CHECK indicator test. Using the DVM (item 2, Table IX) to monitor the output, set the power supply (item 3, Table IX) output to 22.0 + 1 Vdc. Connect the POWER SUPPLY between each point specified–in Table II and GROUND. Rotate the SWITCH FUNCTION CHECK rotary switch through all positions for each connection. Verify that the SWITCH FUNCTION CHECK indicator lamp only “LIGHTS” at the positions specified in Table II.

4.7.3.4 CONTINUITY CHECK indicator test. Using the DVM (item 2, Table IX) to monitor the output, set the power supply (item 3, Table IX) to 22.0 ± 0.1 Vdc. Connect the power supply between each point specified in Table III and ground. Operate CONTINUITY CHECK rotary switch through all positions for each connection. Verify that the CONTINUITY CHECK indicator lamp only “LIGHTS” at the positions specified in Table III.

4.7.3.5 TEST POINT NO. 1 test. Connect the power supply (item 3, Table IX) between TEST POINT NO. 1 (TP1) and GROUND. Using the DVM (item 2, Table IX) to monitor the power supply output voltage, set this voltage to 24.0 + 0.1 Vdc. Set the POTENTIOMETER SELECT rotary switch to the positions of Table IV. Using the DVM, verify the output of 24.0 ± 0.1 Vdc at the connector and pin specified in Table IV for each switch position.

4.7.3.6 TEST POINT NO. 2 test. Connect the power supply (item 3, Table IX) between TEST POINT NO. 2 (TP2) and GROUND. Using the DVM (item 2, Table IX) to monitor the power supply output voltage, set this voltage to 24.0 + 0.1 Vdc. Set the POTENTIOMETER SELECT rotary switch to the positions of Table V. Using the DVM, verify the output of 24.0 ± 0.1 Vdc at the connector and pin specified in Table V for each switch position.

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4.7.3.7 TEST POINT WIPER test. Connect the power supply (item 3, Table IX) between TEST POINT WIPER (TP3) and GROUND. Using the DVM (item 2, Table IX to monitor the power supply output voltage, set this voltage to 24.0 + 0.1 Vdc. Set the POTENTIOMETER SELECT rotary switch to the positions of Table VI. Using the DVM, verify the output of 24.0 + 0.1 Vdc at the connector and pin specified in Table VI for each switch position.

4.7.3.8 RC network setup test. Connect the power supply (item 3, Table IX) between J114-C(+) and J114-A(-). Using the DVM (item 2, Table IX) to monitor the power supply output voltage, set this voltage to 24.0 ± 0.5 Vdc. Connect the resistor (item 5, Table IX) between J22-AY and J22-BL. With the +24 Vdc POWER switch in the "ON" position, verify that the +24 Vdc POWER indicator lamp is "LIT."

4.7.3.9 RC charging test. Set up the conditions of 4.7.3.8. Connect the vertical input of the oscilloscope (item 4, Table IX) between E3(+) and E4. With the oscilloscope set up for internal (+) sweep, verify the waveform of Figure 1 when the POTENTIOMETER SELECT switch is placed in position 7.

4.7.3.10 RC discharging test. Set up the conditions of 4.7.3.8. Connect the vertical input of the oscilloscope (item 4, Table IX) between the E1(+) and E2. With the oscilloscope set up for internal (-) sweep, verify the waveform of Figure 2, when the POTENTIOMETER SELECT switch is rotated from position 7 to position 6. Reconnect the oscilloscope vertical input between J22-AX(+) and J22-BL. Verify the waveform of Figure 2 when the POTENTIOMETER SELECT switch is rotated from position 7 to position 8.

4.7.3.11 Relay test switch. Using the ohmmeter (item 1, Table IX), verify the requirements of 3.5.6.

4.7.3.12 15 V regulator circuit. The 15 V regulator circuit shall be tested as follows in compliance with 3.5.7.

Step 1. Connect power supply (item 3, Table IX) between J114-C(+) and J114-A(-). Set power supply output to +24 V + 0.1 Vdc.

Step 2. Connect DVM between J23-J(+) and TP4(-).

Step 3. Position S3 to ON.

Step 4. Turn the +15 Vdc ADJUST knob to its max CCW position.

Step 5. DVM shall read less than 4.0 V.

Step 6. Turn the +15 Vdc ADJUST knob to its max CW position.

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Step 7. DVM shall read greater than +20.0 Vdc.

Step 8. Connect DVM between J23-f(+) and TP4(-).

Step 9. Repeat steps 4-7.

4.7.3.13 Blocking Diode Continuity. Using the DVM (item 2, Table IX) and the POWER SUPPLY (item 3, Table IX), verify the requirement of 3.5.8.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS9361261.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This test set is designed for drawer mounting into the AN/TSM-100B Electronics System Test Set, which is used for troubleshooting and performance checking of specific weapon system components for the M163A2/M167A2 Vulcan Air Defense System (VADS).

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First article sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of conformance for each lot or shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I , Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Electronics System Test Set AN/TSM-100
Fire Control System
M163A2/M167A2 Vulcan Air Defense System (VADS)

6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publications and Forms Center (NPFC) for publication. This issue is the initial public release of unlimited distribution.

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Army–AR

Preparing activity:
Army-AR

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3 DOCUMENT TITLE

CONTROL ASSEMBLY TEST PANEL TS-4215/TSM-100B: 9361261

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

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